

GREATER CUMBERLAND, MD, USA

WMO: 720355

Lat: 39.617N

Lon: 78.767W

Elev: 236

StdP: 98.52

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03732

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.0	-19.8	0.7	-7.9	-16.9	0.9	-6.1	10.5	-0.1	9.0	1.1	2.2	270	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	33.0	22.3	32.3	22.0	31.1	21.6	24.5	30.2	23.7	29.1	23.0	28.1	3.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.0	26.1	22.4	17.6	25.6	22.1	17.2	25.2	75.1	29.9	72.1	29.2	69.5	27.4	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.2	5.8	DB	-14.7	35.8	2.1	1.8	-16.2	37.1	-17.4	38.1	-18.6	39.1	-20.2	40.4	(4)
(5)			WB	-14.8	25.7	2.5	1.2	-16.6	26.5	-18.0	27.2	-19.4	27.9	-21.2	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	0.1	1.5	6.3	12.3	17.8	21.7	24.0	23.1	19.6	13.3	6.5	2.8		
	DBStd	9.44	4.80	5.48	5.36	4.93	4.42	3.08	2.71	2.67	3.76	4.67	4.42	4.27		
	HDD10.0	1092	308	243	142	31	2	0	0	0	0	18	122	226		
	HDD18.3	2710	564	471	375	189	67	8	1	1	29	171	353	481		
	CDD10.0	1996	2	6	27	100	242	351	434	406	289	119	18	3		
	CDD18.3	572	0	0	1	8	49	108	176	148	69	13	0	0		
	CDH23.3	5338	0	4	13	170	477	987	1666	1263	634	120	4	0		
	CDH26.7	1956	0	0	1	53	148	332	697	466	230	30	0	0		
(14) Wind	WSAvg	1.8	2.1	2.3	2.4	2.3	1.6	1.5	1.4	1.2	1.2	1.6	1.8	1.9		
(15) Precipitation	PrecAvg	927	64	61	85	82	96	85	88	82	86	66	72	67		
	PrecMax	1351	131	124	181	212	185	225	178	204	230	206	191	152		
	PrecMin	675	9	2	19	18	9	21	14	30	2	5	2	16		
	PrecStd	150	32	35	39	44	45	44	47	35	51	45	38	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	22.1	25.2	31.3	32.4	33.8	36.0	34.8	33.7	30.1	22.7	17.5		
		MCWB	11.4	14.9	14.8	16.6	21.1	22.7	23.9	22.2	21.7	20.8	13.8	13.8		
	2%	DB	12.6	17.2	22.2	27.7	30.3	32.2	33.7	32.6	32.0	26.9	19.8	13.0		
		MCWB	8.4	10.9	14.0	16.3	20.0	22.2	23.2	22.3	21.4	19.2	12.8	9.5		
	5%	DB	9.0	13.0	18.7	25.0	27.9	30.3	32.3	31.1	29.8	23.2	16.9	11.1		
		MCWB	6.4	8.4	12.3	15.2	19.2	21.6	22.6	21.8	20.4	16.6	11.2	8.2		
	10%	DB	7.0	9.8	16.1	22.0	26.2	28.0	30.9	29.8	27.3	21.2	14.1	8.6		
		MCWB	4.5	6.1	10.0	13.5	18.4	20.3	22.1	21.5	19.3	16.3	9.6	6.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	15.3	16.9	19.7	23.6	24.5	25.7	25.3	24.0	22.5	16.0	14.4		
		MCDB	17.0	22.8	20.6	26.8	28.5	30.9	32.6	29.8	28.9	26.8	21.6	17.3		
	2%	WB	9.3	11.7	15.1	17.7	22.2	23.6	24.7	24.2	22.8	20.7	14.1	11.3		
		MCDB	12.2	15.3	20.1	23.7	26.8	29.3	31.1	28.8	28.3	24.3	17.8	12.9		
	5%	WB	6.7	8.9	13.3	16.3	21.0	22.8	23.9	23.3	22.0	18.9	12.0	8.1		
		MCDB	8.6	13.7	18.2	22.1	25.0	28.0	29.7	27.8	26.9	22.8	15.3	9.7		
	10%	WB	4.8	6.3	10.8	14.9	19.9	22.0	23.2	22.5	20.9	17.1	10.2	6.6		
		MCDB	6.8	9.6	14.8	20.9	24.3	26.7	28.4	26.9	24.9	19.7	13.5	8.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.9	10.6	12.6	12.2	11.7	12.1	11.8	12.3	11.3	10.2	7.5		
		MCDBR	12.1	14.9	16.8	18.8	15.7	14.1	14.2	14.2	15.9	15.4	15.1	11.7		
	5% WB	MCWBR	8.8	9.7	9.6	8.7	6.7	5.6	4.8	5.0	6.1	7.6	9.1	8.5		
		MCWBR	11.1	14.3	14.9	15.2	12.8	11.9	11.9	10.9	12.8	12.5	12.9	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.322	0.351	0.393	0.433	0.493	0.518	0.516	0.424	0.366	0.323	0.306		
	taud		2.453	2.384	2.384	2.297	2.202	2.063	2.010	1.995	2.239	2.406	2.495	2.486		
	Ebn at Noon		862	897	904	882	850	796	771	760	815	834	837	837		
	Edh at Noon		80	98	109	127	143	165	172	171	125	96	77	72		
(44) All-Sky Solar Radiation	RadAvg		1.78	2.62	3.57	4.66	5.20	5.77	5.67	5.11	4.20	3.02	2.12	1.53		
	RadStd		0.17	0.36	0.23	0.41	0.49	0.45	0.34	0.29	0.53	0.35	0.22	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	+60

Nomenclature: See separate page